

Project 3: SQL Data Analysis Milestone Report

DecodeLabs Industrial Data Analytics Training Kit — Batch 2026

Domain:	Data Infrastructure & Analytics	Date:	June 13, 2026
Milestone:	Project 3 (Extraction Phase)	Status:	Verified & Complete

1. Executive Summary

Yeh report DecodeLabs Industrial Training ke Project 3 (SQL Data Analysis) ka official documentation hai. Is project ka basic objective spreadsheet viewing ke traditional methods se shift hokar core relational database logic ko implemented karna hai. E-commerce dataset ka use karte hue high-value transactions isolate kiye gaye hain, pattern matching algorithms execute kiye gaye hain aur granular sales performance ko summary insights mein transform kiya gaya hai.

2. Core Technical Architecture & Database Engine Mechanics

SQL script run karne se pehle, database management engine query execution ko prioritize karne ke liye teen essential phases execute karta hai:

- **Parsing & Binding:** SQL parser structural syntax ki testing aur accuracy verify karta hai taake grammatical consistency maintain rahe.
- **Cost-Based Optimizer (CBO):** CBO engine computer memory, CPU utilization aur storage I/O cost evaluate karke pin-pointed data fetch karne ka shortest execution path compute karta hai.
- **Execution Plan:** Final processed optimization path ke mutabiq relational data tables se data blocks ko memory buffer cache mein stream kiya jata hai.

3. Core Project Requirements & Structural Queries

Milestone parameters ke mutabiq target dataset par operational components map kiye gaye hain:

Requirement 1: Advanced Filtering Strategy (WHERE Funnel)

Is condition ka basic criteria high-volume items extraction segment ko isolate karna hai jahan transactional items `Quantity > 3` ho aur unka overall financial value basket size `TotalPrice > $500.00` cross kare.

Requirement 2: Pattern Matching Engine (LIKE Operator)

Strategic discount campaigns track karne ke liye marketing identifier validation algorithms implement kiye gaye hain, jo exclusive `WINTER%` pattern mapping filters execute karte hain.

Requirement 3: Categorical Bucketing (GROUP BY & Multi-Variable Aggregation)

Bikhre hue flat structural orders ko compact summary arrays mein map karne ke liye multi-variable aggregations calculate kiye gaye hain (COUNT , SUM , aur AVG).

4. Sample Executive Data Summary Insights

Dataset insights compilation ke bad logical metrics structure neeche diye gaye analytical layout ke mutabiq observe kiya gaya hai:

Product Category	Total Orders	Total Units Sold	Average Unit Price (\$)	Total Revenue Generated (\$)
Desk	340	1,120	450.50	504,560.00
Monitor	295	980	320.15	313,747.00
Printer	210	640	290.00	185,600.00
Tablet	185	590	380.40	224,436.00

Strategic Deployment Note: GitHub repository par code evaluation transparent rakhne ke liye complete database constraints verification aur logical sorting optimization script fail-safe structural layer ke sath ensure kar di gayi hai.

5. Repository Execution Guide

1. Apne RDBMS console (MySQL Workbench / PostgreSQL pgAdmin) mein database initiate karein.
2. Provided table definition schema block execute karke database environment setup karein.
3. Main analytic logic execution blocks run karein aur insights arrays dynamically pipeline karke output screen verify karein.